

C++ Concurrency Tools Comparison Chart

Tool	Purpose	Header	Features
std::thread	Create and manage threads	<thread>	Basic threading, join, detach
std::mutex	Mutual exclusion	<mutex>	lock, try_lock, unlock
std::recursive_mutex	Reentrant locking	<mutex>	Allows same thread to lock multiple times
std::timed_mutex	Timed locking	<mutex>	try_lock_for, try_lock_until
std::shared_mutex	Multiple readers, single writer	<shared_mutex>	lock_shared, unlock_shared
std::condition_variable	Thread synchronization	<condition_variable>	wait, notify_one, notify_all
std::future & std::promise	Asynchronous result retrieval	<future>	get, set_value, wait
std::async	Run task asynchronously	<future>	launch::async, launch::deferred
std::atomic	Atomic operations	<atomic>	lock-free operations, memory_order
std::lock_guard	RAII-style mutex wrapper	<mutex>	Locks mutex on construction, unlocks on destruction
std::unique_lock	Flexible mutex management	<mutex>	defer_lock, try_lock, timed locking